Research Statement

By Yi Ping

I study asset pricing and macroprudential policy using general equilibrium methods with endogenous leverage and heterogeneous agents. My research examines how the feedback loop between household leverage, mortgage rates and spreads, and housing prices drives boom-and-bust housing cycles, as well as the aggregate and distributional effects of leverage-related financial regulations. In addition to household markets, I study institutional risk management, where I examine how currency hedging and endogenous repo haircut drive safe-asset fire sales.

I. Leverage cycles and household life cycles

My job market paper, "**Leverage Cycles over the Household Life Cycle: A Quantitative Model of Endogenous Leverage and Default**," provides a framework to address two central questions in macro-finance: How do households' endogenous leverage choices over the life cycle drive boom-and-bust cycles in housing prices? And who ultimately bears the cost of a housing recession?

Addressing the first question requires endogenous leverage determination, while addressing the second requires sufficiently rich agent heterogeneity to capture distinct household portfolios and risk exposures. The theoretical literature on endogenous leverage, pioneered by Geanakoplos (1997), Geanakoplos and Zame (2014), and Fostel and Geanakoplos (2008, 2014, 2015), establishes a strong link between leverage and collateral asset prices and shows how leverage amplifies asset price volatility. Yet, because pricing a continuum of contracts and determining the optimal ones are both conceptually and computationally demanding, existing quantitative models feature limited agent heterogeneity. A central goal of my research agenda is to expand the set of questions that endogenous leverage theory can address by advancing it to the quantitative frontier with rich household heterogeneity.

To do so, I develop a general equilibrium model with 30 overlapping generations in which risk-averse households trade housing and a continuum of non-recourse mortgage contracts collateralized by housing. Households default on mortgages when outstanding debt exceeds the value of the collateral. Equilibrium prices of the contracts trace out an endogenous *Credit Surface* that maps each loan-to-value (LTV) ratio to an interest rate, where higher-LTV loans carry higher default risk premiums and hence higher interest rates.

The model delivers three main results. First, leverage progressively declines with age. Facing a growing income profile, young households leverage heavily against housing, exhausting their collateral capacity to issue risky contracts on which they default in a recession. Subsequent cohorts of young borrowers start diversifying their portfolios by issuing both risky and riskless contracts and are unconstrained. Middle-aged and older households lend to the

young. Over the life cycle, households adjust their portfolios from positions highly exposed to aggregate risk toward more balanced positions.

Second, in a recession, housing prices and leverage plummet simultaneously while mortgage rates and spreads surge. Falling house prices push young borrowers underwater and trigger defaults. Lenders bear default losses and also see lower housing wealth, and their marginal utility of consumption increases. Consequently, lenders demand higher returns and a larger risk premium, causing the riskless rate to rise and mortgage spreads to spike. Because costlier risky loans reduce the borrowing capacity of housing, its collateral value collapses, amplifying the drop in housing prices.

Third, unconstrained borrowers and lenders bear the brunt of a recession, whereas constrained young borrowers are shielded by non-recourse default. Although constrained young borrowers are the most leveraged, they can offload losses to lenders by walking away from underwater loans. This leaves them with mild consumption losses while allowing them to purchase more housing at depressed prices, albeit with lower leverage. Instead, severe consumption losses fall on unconstrained young borrowers, whose residual home equity is most exposed to aggregate risk, and on lenders, who absorb both housing wealth declines and default losses.

II. Monitoring the Credit Surface and macroprudential policy

A key policy implication of my job market paper is that policymakers should monitor the full spectrum of the Credit Surface, not just low-LTV loans with riskless interest rates. Due to a spike in the default risk premium during a recession, risky loans become disproportionately more expensive, exacerbating the crash in housing prices. Policies that target only riskless loans can therefore fail to prevent such a crash.

Building on the framework of my job market paper, my ongoing project, "**Monitoring the Credit Surface: The Effect of Mortgage Tax and Default Tax**" (joint with Ana Fostel and Eric Young), studies the design of policies that regulate mortgage leverage and default. Existing macroprudential studies evaluate mortgage market interventions under exogenous LTV caps, which shuts down the feedback loop between leverage and asset prices. We aim to fill this gap by examining how households respond to a certain type of intervention by adjusting their portfolios, and in aggregate how their responses feed back into general equilibrium price adjustments. Specifically, we examine the aggregate and distributional effects of two broad classes of interventions: mortgage taxes and subsidies imposed in normal times, and default penalties combined with compensation to lenders in downturns.

The first class of interventions aims to curb excessive leverage ex ante in normal times. We consider four regimes along two dimensions: (i) whether the policy targets only risky loans or both risky and riskless loans, and (ii) whether it taxes borrowers exclusively or simultaneously taxes borrowers and subsidizes lenders at the same rate. In each regime, the government uses agent-specific lump-sum transfers that exactly offset the taxes paid by

borrowers or the subsidies received by lenders, in order to eliminate direct wealth effects. In the second class of interventions, we consider compensating lenders in case of default, and funding the compensation by taxing defaulters' consumption in proportion to the volume of contracts on which they default.

We first analyze a two-period, four-agent version of the economy. This framework highlights why endogenous leverage in general equilibrium and granular heterogeneity are both essential for policy evaluation. First, with both lenders and borrowers determining interest rates across the menu of loans, policy interventions can have opposite effects compared to models with outside lenders. We show that taxing borrowers does not necessarily hurt them, and subsidizing lenders does not always help them, either. When a tax on borrowers is paired with a subsidy to lenders, lenders expand credit supply as they see a higher return on lending with the subsidy, bidding up mortgage contract prices and lowering interest rates. Because the collateral constraint restricts the quantity of contracts issued to the physical housing stock, higher contract prices allow borrowers to raise more funds per unit of pledged housing. This increases the collateral value of housing and delivers a first-order welfare gain for borrowers while hurting lenders. Second, our four-agent structure demonstrates the importance of distinguishing between constrained and unconstrained borrowers. Under a range of mortgage tax rates without subsidizing lenders, constrained borrowers have no room to adjust their portfolios in response to the tax, and they continue issuing risky debt exclusively to meet their liquidity needs. Unconstrained borrowers, by contrast, have room to freely reshuffle their portfolios between risky and riskless contracts, allowing them to achieve a small welfare gain. In addition, taxing only constrained borrowers and only unconstrained borrowers push housing prices to move in opposite directions.

Building on these insights, we extend our analysis to the full 30-generation quantitative model and search for the optimal policy for each type of intervention. We have solved for the stationary equilibria across a grid of policy parameters and approximated generation-specific value functions. We construct a social welfare function defined as the sum of the remaining lifetime utilities of all currently living generations and the discounted lifetime utilities of future unborn generations. Using this welfare metric, we are currently refining the policy parameter grid to locate the optimal policy rule across the candidate regimes.

III. The short-term liquidity cost of long-term solvency hedging

Beyond household-based asset pricing, I am also interested in topics of institutional risk management through the lens of general equilibrium models with endogenous leverage. Motivated by recent institutional liquidity crises[1], my ongoing solo-authored project, "**The**

[1]Examples include the March 2020 "dash for cash" (Czech et al., forthcoming), the 2022 UK gilt crisis among liability-driven investment (LDI) pension funds (Bank of England, 2022; Chen and Kemp, 2023; Pinter, Siriwardane and Walker, forthcoming), and similar liquidity squeezes among European pension funds (Jansen et al., 2026).

Liquidity Cost of Currency Hedging," studies how the maturity mismatch between long-term assets and short-term hedges exposes institutions to liquidity distress and the subsequent tightening of repo margins.

Pension funds and insurers commonly hold long-term foreign-currency assets (typically in dollars) while mandated to meet long-term liabilities in local currency. To ensure their long-term solvency, they use short-term forwards or swaps to lock in future exchange rates. Because these hedge contracts typically mature in the short term compared to the foreign-currency assets, they must be rolled over at each interim date before the assets mature. At an interim date, if the dollar appreciates in the spot market, the local-currency value of the long-term foreign-currency assets rises, but investors must immediately raise cash to meet variation margin calls, a phenomenon coined "Liquidity After Solvency Hedging" (LASH) risk by Alfaro et al. (2024).

IV. The short-term liquidity cost of long-term solvency hedging

Beyond household-based asset pricing, I am also interested in topics of institutional risk management through the lens of general equilibrium models with endogenous collateral capacity. Motivated by recent institutional liquidity crises[2], my ongoing solo-authored project, "**The Liquidity Cost of Currency Hedging**," studies how the maturity mismatch between long-term assets and short-term hedges exposes institutions to liquidity distress and endogenous contractions in repo borrowing capacity.

Pension funds and insurers commonly hold long-term foreign assets (typically denominated in dollars) while mandated to meet long-term liabilities in local currency. To ensure they are solvent when the foreign assets mature, they use short-term forwards or swaps to lock in future exchange rates. Because the hedge contracts have shorter maturities than the foreign assets, they must be rolled over at each interim date before the assets mature. At an interim date, if the dollar appreciates in the spot market, the local-currency value of the long-term dollar assets rises, which improves solvency, but creates a variation margin for which they must immediately raise cash to keep hedging the long-term solvency risk, a phenomenon coined "Liquidity After Solvency Hedging" (LASH) risk by Alfaro et al. (2024).

To raise cash at an interim date, institutions use their holdings of domestic government bonds as collateral to borrow in the repo market. However, repo haircuts spike during the crisis, which forces institutions to sell their bond collateral directly and further pushes down the bond prices. Existing theoretical papers study hedging under partial equilibrium borrowing constraints (Mello and Parsons, 2000; Deep, 2002; Rampini, Sufi and Viswanathan, 2014) or variation margins with fire sales (Biais, Heider and Hoerova, 2021), but there is

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not yet a general equilibrium model that links repeated hedge rollovers and endogenous repo margins to rationalize the crash of collateral prices.

I build a general equilibrium model to examine this mechanism. By the binomial no-default theorem in Fostel and Geanakoplos (2008, 2015), repo borrowing capacity today is determined by the worst-case price of the bond collateral in the next period. In the state where an adverse exchange-rate shock triggers a liquidity crisis for funds, lenders believe that funds will be forced into a next-period fire sale and that future collateral prices will crash. Driven by this belief, lenders raise the repo haircut, which in turn forces funds to sell bonds to raise cash, and drives down the bond price even further.

This model has the potential to answer a set of policy questions regarding non-bank financial stability. The research topic I have in the pipeline is the following: when institutions face a liquidity crisis, central banks often respond by injecting cash into the repo market. When repo haircuts are driven by lenders' beliefs about future fire sales, how effective is injecting cash into the repo market compared to directly buying bonds to support collateral prices?

In my future research, I am committed to advancing our understanding of asset pricing and financial policy using general equilibrium models with endogenous leverage. Thanks to recent rapid developments in machine learning, it is now possible to solve high-dimensional, complex models. I plan to keep exploring heterogeneous agents and realistic market frictions in my work.

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